NGOC MINH TRAN (Irish & Vietnamese)

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I have been holding a director-level position at Mastercard Ireland with expertise in AI/ML, focusing on GenAI. Before that I was a staff research engineer with IBM Research Lab Ireland working on AI Security, AutoAI and AI Federated Learning. I also had experience with a US startup company in Silicon Valley working on AI and Big Data, after I finished my PhD and Postdoc in ML. Altogether, I have 15+ years experience in AI domains. Links within the CV are track records.

EDUCATION

- Postdoc, Institute for Research in Computer Science and Automation (INRIA), France 2012 – 2013
- PhD. in Computer Science, Leiden University, The Netherlands 2008 – 2012

ACHIEVEMENTS

Awards

- IBM Research Division Award in **Security AI** 2019.
- IBM Research Division Award in **Auto AI** 2019.
- IBM Outstanding Technical Achievement Award in **Security AI** 2019.
- IBM Outstanding Technical Achievement Award in **Federated AI for Hybrid Cloud and Edge** 2021.
- IBM First Plateau Invention Achievement Award 2019.
- IBM Second Plateau Invention Achievement Award 2019.
- IBM Third Plateau Invention Achievement Award 2020.

Open Source Recognitions

- Open Source Strategic Committer in 2019.
- Pull-Shark achievements 2 times in Github.
- Arctic Code Vault contributor in Github.

Publications and Patents

- First author (10) and co-author (6) of 16 scientific papers in artificial intelligence (AI) and clouds.
- Lead inventor (12) and co-inventor (9) of 21 US patents in security AI, federated AI, auto AI, generative AI and Fintech AI.
- See Google Scholar for the full list of publications and patents.

Professional Certificates

- Developer Certification for Apache Spark, certified by Databricks and O'Reilly Media, Jan 2016.
- Oracle Certified Professional, Java SE 6 Programmer, Oct 2015.
- Revolution R Enterprise Certified Specialist, Aug 2015.
- Cloudera Certified Specialist in Apache HBase, Dec 2014.
- Cloudera Certified Developer for Apache Hadoop, Oct 2014.
- Cloudera Certified Administrator for Apache Hadoop, Oct 2014.

TECHNICAL AND PROJECT MANAGEMENT SKILLS

• AI, machine learning, deep learning, genAI, LLM	Very good
• Python	Very good
• Hadoop, PySpark	Good
• Tensorflow, Pytorch, MLOps, Mlflow, Docker, Databricks	Good
• Agile Scrum, Jira, Git	Experienced
• Kubernetes, AWS	Knowledgeable

PROFESSIONAL EXPERIENCE

Principal AI Engineer at Mastercard Ireland (5 years) Oct 2021 – Current

- **Foundation Model Architecture:** Architect and build a proprietary transaction-based Foundation Model using Transformer architectures, scaling deep learning training across hundreds of billions of global transactions utilizing multi-node H100 GPU clusters.
- **GenAI & RAG Implementation:** Design and implement robust RAG systems and fine-tune foundational LLMs on internal enterprise data to accelerate cross-functional project onboarding.
- **Banking & Commercial Use Cases:** Develop ML applications directly driving core business metrics, including SME Credit Risk Scores, Hidden SME Identification, Fraud Detection, Customer Attrition/Reactivation, and Liquidity Forecasting.
- **Financial Optimization:** Optimize multi-million dollar compute budgets and MLOps pipelines on Databricks for ultra-large-scale foundational model deployment.

Staff Research Engineer at IBM Research Lab Ireland (4 years and 9 months) Feb 2017 – Oct 2021

- Be a key member of the open source AI project Adversarial Robustness Toolbox (ART), play the role of architect, maintainer, reviewer and contributor with 1355 commits, 73423++ and 45949-- of lines of codes. ART is supported by the US Defense Advanced Research Projects Agency (DARPA), contract number HR001120C0013. **ART has received 6300+ stars, 1300+ forks on Github and has been cited by 1000+ research papers.**
- Contribute to the key AI project of IBM, the Federated Learning (IBMFL). IBMFL is a Python framework for federated learning in an enterprise environment. It has 2 versions: an enterprise version and an open source version, **which has received 530+ stars, 140+ forks on Github and has been cited by 250+ research papers**.
- Lead research projects for papers and patents in AI such as Automation of Image Data Preprocessing (AutoDP).
- Contribute to the key AI project of IBM, the One Button Machine, which belongs to the AutoAI domain and is implemented as a Watson Machine Learning service of IBM Watson Studio.

Senior Data Scientist at ARIMO (Panasonic, Remote) 2016 – 2017

ARIMO is a California-based leader in deep learning and AI, ranked among the ten most innovative startup companies in data science in 2016. My work includes developing IoT-centric AI models for commercial and manufacturing applications.

Solution Architect (Data Science) at DirecTV (AT&T, Remote) 2014 – 2016

- Manage and coach an R&D team of junior data scientists.
- Data science tech lead of big data project, which aims at building a cloud-based recommendation engine to recommend TV/movies/sport programs to ~40 millions end users of DirecTV.

Postdoc at INRIA, France 2012 – 2013

Involve in an EU-FP7 project, which aims at incorporating innovative hardware and network technologies seamlessly into data centers that provide PaaS cloud infrastructures.

PhD candidate at Leiden University, The Netherlands 2008 – 2012

My PhD mainly focuses on statistical data analysis, modeling and machine learning, especially with stochastic random processes and long-term time series in the domain of clusters and clouds.